

Day 1 — What is the Sun Doing?

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will describe the Sun’s daily and annual apparent motion and articulate the question: *does the Sun rise from exactly the same spot every day?*

Materials

- Whiteboard
- A 30 cm wooden dowel or pencil + a piece of cardboard / paper to plant it on (the gnomon)
- A compass (or a phone with a compass app)
- A printout of a year-calendar (Jan to Dec, all dates visible) — one per pair
- Optional: a photo of the room taken at the same time on different days (a teacher prep item)

Vocabulary introduced today

- *Sūrya* (IAST: *sūrya*; lit. “the Sun”) — the luminary at the centre of our solar system; the object we will track for the next 10 days.
- *krānti-vṛtta* (IAST: *krānti-vṛtta*; lit. “deflection-circle”) — the **ecliptic**, the apparent path the Sun traces across the sky over a year.

Lesson flow

Warm-up (5 min)

- Prompt on board: “*Where was the Sun this morning when you left home?*”
- Take 3–4 responses. Most will say “east” or “in front of me / behind me.” Some will say “I didn’t notice.” That’s the honest answer — and the launching point.
- Teacher: “*Today we’re going to start noticing. For two weeks we’ll track the Sun. The Indian astronomical tradition has been doing exactly this for 2000+ years.*”

Core (20 min)

1. **Two motions, one Sun.** Sketch on the board:
 - **Daily motion:** rises east-ish, climbs, sets west-ish. Takes ~24 hours.
 - **Annual motion:** the rising spot shifts north (summer) and south (winter) along the horizon, completing a full cycle in ~365.25 days.
2. **The Sun’s apparent path on the sky.** Show this:

NORTH
|
W---+---E ← in March/September the Sun rises due east

| and sets due west
SOUTH

W---+---E ← in June the rising point shifts NORTH of east
(high arc in the sky, long days)

W---+---E ← in December it shifts SOUTH of east
(low arc, short days)

- The path the Sun traces against the *stars* over a year is called the **ecliptic** (*krānti-vṛtta*).
- From Earth, we don't see the stars in daytime — but Indian astronomers worked out the ecliptic by noting which stars came up just before sunrise and just after sunset.

3. **Two questions for the module.** Write on the board:

- **Q1:** Why does the rising spot shift? (*Answer = axial tilt — Day 2.*)
- **Q2:** What did Indian astronomers call the Sun's slow north-south swing? (*Answer = uttarāyaṇa and dakṣiṇāyana — Day 3.*)

Activity (15 min) — Shadow-stick observation

If the weather permits and the schoolyard has a sunny spot:

- Plant the gnomon (the dowel) upright in a piece of cardboard. Mark north on the cardboard using the compass.
- At a known time (say, the start of the lesson), trace the shadow of the gnomon onto the cardboard. Write the time and the date next to the line.
- Return to the same gnomon in 10 minutes. The shadow has moved. Trace the new line.
- Discuss: “*The Sun seemed to stand still while you were in class — but the shadow moved. What does that tell you about how the Sun moved?*”

If weather doesn't permit: do the activity indoors with a strong torch (acting as the Sun) and a pencil-on-paper gnomon. Move the torch slowly along an arc to mimic the daily motion. Each “minute” of torch travel = ~4 minutes of real Sun travel.

Safety reminder: the students look at the **shadow**, not the Sun. Anyone caught glancing up at the Sun pays a forfeit (a fun one — like doing the daily prompt next class).

Wrap-up (5 min)

- Exit prompt: “*In one sentence — how can you tell tomorrow morning whether the Sun has ‘moved north’ or ‘moved south’ since today?*”
- Take 2 responses. Acceptable answer: “*Compare the sunrise spot to a known landmark on the horizon — has the rising point moved closer to north or to south?*”
- Preview Day 2: “*Tomorrow — WHY does the Sun's spot shift? Bring a globe if you have a small one at home.*”

Student-facing content

The Sun has two motions you can see:

- **Daily** — rises east-ish, sets west-ish. Takes 24 hours.
- **Annual** — the rising spot drifts north and south along the horizon over a year.

The Sun's yearly path across the stars is called the **ecliptic** (*krānti-vṛtta*). Indian astronomers, like astronomers everywhere, traced this path by watching which stars were just visible at dawn and dusk.

Today, you have noticed the daily motion (via the gnomon). Over the next two weeks, you'll learn the names for the annual motion: *uttarāyaṇa* (northward course) and *dakṣiṇāyaṇa* (southward course).

Homework

- For three mornings this week, when you leave home, glance at the eastern horizon. Note where exactly the Sun rises relative to a landmark (a tree, a building, a flagpole). Sketch the spot on a small map — three sketches.
- Bring the calendar printout to Day 2.

Differentiation

- **One tier up:** Find out (online or in the school library) the **time** of sunrise and sunset for today AND for one day exactly six months from now. By how many minutes does the day length differ?
- **One tier down:** Just do the morning-glance — don't worry about sketching maps. Three mental notes are enough.

Teacher notes

- Some students will know the Sun-rises-in-the-east rule and stop there. Push: “Always east? Or east-ish?” The “ish” is the entire module.
- Don't introduce axial tilt yet. That's Day 2. Let students sit with the *observation* of the shifting spot before getting the *explanation*.
- The shadow activity is more valuable than the explanation. If you only have 30 minutes, run the activity and trim the Core.